Comprehensive Cloud Solution Design

Deepika Agnihotri

University of Maryland Global Campus

CLCS 605: Introduction to Cloud Computing

Professor Anthony Ayodele

Scenario

A small e-commerce company has recently migrated their website and database to the cloud. They need a backup strategy and monitoring solution to ensure business continuity and optimal performance.

Backup Strategy

1. Backup Frequency
Selection: Daily
Explanation: Daily backups help to reduce the risk of losing important data by ensuring that information is captured at least every 24 hours.

2. Backup Storage Solution
Selection: AWSS3
Explanation: Selected AWS S3 because it offers a highly cost-effective storage solution that is designed for exceptional durability and security, making it well-suited for storing critical backup data. It allows organizations to automatically transition or archive data over time. These features make AWS S3 an ideal choice for handling frequent backups in a dependable and effective manner.
3.RecoveryTimeObjectives (RTO)
However, for a small e-commerce company, an outage means direct loss of sales and potential reputational damage. Therefore, fast recovery time is crucial for business continuity.
Selection: 4 hours
Explanation: This aligns with the maximum acceptable downtime established by the business, making sure that operations can resume within the time limit the organization considers manageable. At the same time, it remains realistic and fully achievable using cloud-based restoration capabilities, which provide the flexibility and speed needed to meet these recovery expectations.
4. Recovery Point Objective (RPO)
Selection:1hour
Explanation: This ensures that a high-transaction e-commerce platform suffers minimal data loss, even with continuous activity throughout the day. By protecting the latest information, it supports real-time order processing, customer interactions, and other critical operations that depend on up-to-date data.

Disaster Recovery Plan
The disaster recovery plan uses automated snapshots, S3 backups, and cross-region replication to keep data available during major outages. A pre-provisioned standby environment allows compute and database replicas to scale up quickly in another region within the RTO. RDS automated backups, AWS Backup, and Route 53 health checks support automatic traffic redirection. DR tests will confirm accurate restoration and failover readiness.

Monitoring Framework

1. Key Metrics to Monitor
CPU Usage
Explanation: Signifies computing bottlenecks during peak traffic and prevents performance degradation.
Database Query Performance
Explanation: Making sure of database responsiveness for real-time transactions and reducing cart giving up.
Website Uptime
Explanation: Critical for 24/7 operations to detect outages immediately.
Error Rates
Explanation: Helps detect API issues, or deployment problems early.

2. Monitoring Tools
Selection: Datadog
Explanation: Provides unified monitoring across infrastructure, applications, databases, and logs with easy alert configuration and dashboards.

3. Alert Thresholds
Metric: CPU Usage
Threshold: >80% for 5 minutes
Explanation: Helps prevent sustained overload that may lead to degraded performance during traffic spikes.
Metric: Database Query Performance
Threshold: Average query time >200 ms
Explanation: Signals latency issues that can affect search, checkout, or product page responsiveness.
Metric: Website Uptime
Threshold: Any downtime >1 minute
Explanation: Ensures immediate alerts for quick response to minimize potential revenue loss.
Metric (Optional): Error Rates
Threshold: >2% 5xx errors for 3 minutes
Explanation: Alerts on early signs of backend failures or service disruptions.
Metric (Optional): Network Latency
Threshold: >250 ms
Explanation: Indicates a slow customer experience caused by network congestion.

Cost Optimization
To cut costs, the plan relies on S3 Standard storage with lifecycle rules that move older backups

into S3 Glacier tiers. Centralizing monitoring avoids having too many different tools and reduces the cost of software licenses. Autoscaling is set to expand only during traffic spikes, reducing compute usage during regular load time. The DR setup uses a warm standby rather than a fully active-active replication model, balancing reliability with budget limitations.

Justification Document

A small e-commerce business serving 10,000+ customers each day requires a cloud operations plan that balances trustworthiness, cost savings, and performance. The backup, disaster recovery, and monitoring setups are designed to keep systems robust while respecting the company's budget and compliance obligations.

Backup Strategy

Daily backups were chosen because they minimize potential data loss without significantly increasing storage costs. For an e-commerce platform where transactions occur continuously, losing more than a day's data could result in financial and customer trust impacts. AWS S3 was chosen for backup storage due to its high durability, cost-efficiency, and built-in encryption options that satisfy regulatory requirements for customer data protection. It integrates with AWS Backup and RDS snapshot features, prepare the backup workflow easier to automate and maintain.

The Recovery Time Objective (RTO) was selected at 4 hours to align with the business's stated maximum acceptable downtime. This time limit is achievable with a warm-standby architecture and avoids the excessive cost associated with fully active-active deployments. The Recovery Point Objective (RPO) of one hour is critical for avoiding data loss during outages. Because the platform processes lots of real-time transactions, losing multiple hours of transaction data could result in lost orders and customer service issues.

Disaster Recovery Plan

The disaster recovery (DR) plan leverages cross-region replication and automated snapshots to keep an up-to-date copy of the environment. A warm-standby setup is cost-effective, maintaining essential infrastructure partially active instead of running a full secondary environment. During a primary region outage, the standby can scale to production within the RTO. Route 53 health checks enable automatic DNS failover for quick traffic routing. Regular testing ensures backups are restorable, and failover procedures remain reliable as the system evolves.

Monitoring Framework

Monitoring plays a critical role in maintaining site performance and ensuring that issues are detected promptly. Tracking CPU usage helps identify compute saturation during peak periods, such as holiday seasons. Monitoring database query performance is equally important, as slow queries can negatively impact search results, conversion rates, and the checkout process. Website uptime is essential for continuous business availability because even brief outages can lead to

revenue loss. Additionally, monitoring error rates provides insight into backend failures, integration issues, or faulty deployments, enabling quick intervention before customers are affected.

Datadog was selected as the monitoring solution because it consolidates infrastructure, application, and log monitoring into a single platform. This integration reduces operational complexity, offers clear insights through customizable dashboards, and seamlessly integrates with cloud-native services to support efficient alerting and troubleshooting.
Alert thresholds were chosen to detect faults early while preventing alert fatigue. CPU over 80% for extended periods may indicate swamped instances. Database response times above 200ms begin to degrade customer experience. Any site downtime warrants immediate attention, and small spikes in 5xx errors often signal breaking changes or infrastructure issues. Together, these thresholds create a comprehensive and responsive alerting system.

Cost Optimization
Cost considerations influence each component of the strategy. S3 lifecycle policies transition older backups to cheaper archival storage while maintaining compliance and retention requirements. Compute usage is optimized through autoscaling policies that expand only during traffic surges, preventing unnecessary resource allocation. The warm-standby DR architecture reduces ongoing operational costs compared to a fully active-active solution. Consolidating monitoring into a single tool further reduces licensing and administration expenses. Overall, the strategy delivers business continuity while respecting the company’s budget constraints.

Simple Cost-Benefit Analysis:

Options	Cost Impact	Benefits	Key point
Daily Backups + S3 (Recommended)	Low–Moderate	High durability, compliance support, automated workflows	Most cost-effective for required protection
Active-Active DR	Very High	Fastest failover, zero downtime	Too expensive for budget; exceeds needs
Warm Standby (Recommended)	Moderate	Meets 4-hour RTO, high availability	Best balance of cost and resilience
Datadog Monitoring	Moderate	Unified visibility, fewer tools, faster troubleshooting	Reduces complexity vs multiple separate tools
Multiple Monitoring Tools	High	Redundant visibility	Increases operational overhead

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